Lab test

Rules, format, task types

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Lab objectives

Prove three skills, once

Solve three tasks from a one-page spec in one hour: a device message, an offline write, and a remote read.

Worth 15 points

5 points per task, graded at the end of the test.

Time
1 hour

Submit
To the lab instructor, when the time is up

Rules

1. You work alone. No help from anyone else, no copying a classmate's work.
2. One hour, one sitting, starting when you receive the spec.
3. You must be able to explain any line the grader points to, in your own words.

The clock does not pause. The hour is not extended for any reason.

Test format

One page, handed at the start

Three tasks. Nothing changes once the clock starts.

One hour, then you submit

Work until time is up, then submit your answers to the lab instructor. No extensions.

Every spec has the same three kinds of task, described next.

Task types

Publish or subscribe over MQTT

Talk to the broker: publish a reading, or subscribe and show a value live, with reconnect handled.

One offline-capable write

A local write goes through an outbox and reaches the server once connectivity returns.

One gRPC or BLE read

A unary call to the Go server, or a characteristic read from a peripheral, parsed into a model.

Lab test

Three tasks · one hour